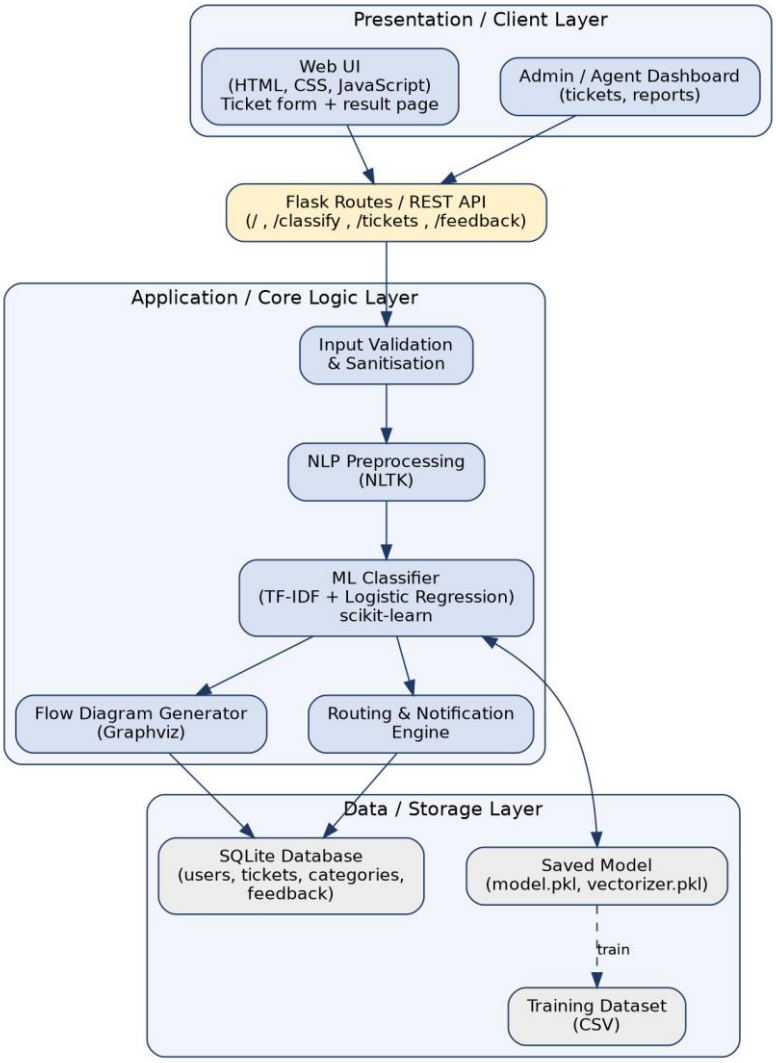
Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-3818
Project Name	Auto_Ticket_Classification_Using_Flow_Diagram
Maximum Marks	5 Marks

Solution Architecture Diagram

The solution follows a three-layer architecture: Presentation Layer, Application (Core Logic) Layer and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Web pages where users raise tickets and see results, and where agents/admin view tickets and reports.	HTML, CSS, JavaScript, Jinja2, Chart.js
API Gateway (Flask Routes)	Receives requests from the browser, calls the core logic and returns JSON or HTML responses.	Flask, REST endpoints

Component Name	Description / Role in Architecture	Technologies Used
Input Validation & Preprocessing	Checks input, cleans text (lowercase, stop-words, lemmatisation).	Python, NLTK
ML Classifier	Converts text to TF-IDF features and predicts category and confidence; saved model is loaded at start-up.	scikit-learn, joblib
Flow Diagram Generator	Builds a diagram of each ticket's journey (received, cleaned, classified, routed).	Graphviz
Routing & Notification Engine	Maps category to support team, updates status and notifies user/agent.	Python, SMTP (optional)
Database	Stores users, tickets, categories, predictions and feedback.	SQLite
Model & Dataset Storage	Stores the trained model, vectoriser and training dataset.	joblib files, CSV